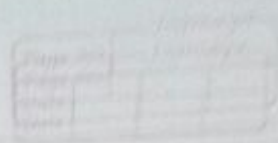


Classmate
8575



5/10/22

- A well defined unordered collection of distinct elements is called a set.

$$\{1, 1, 2\} = \{1, 2\} = \{2, 1\}$$

- A set with no elements denoted as ϕ or $\{\}$ is called null set / empty set.

$\{\phi\} \rightarrow$ set containing ϕ as element

① Roster / Tabular form:-

② Set Builder Form - Notation:-

$$\{a, b, c\}$$

$$A = \{a, e, i, o, u\}$$

$$A = \{x : x \text{ is a vowel}\}$$

16/09/22

Classmate

Topic:- Set, powerset, operation on set

$N \rightarrow$ set of all natural numbers $1, 2, 3, 4, \dots$

$Z \rightarrow$ set of all integers

$Z^+ \rightarrow$ set of all positive integers. 0 whole no.
0 नहीं आता

$Q \rightarrow$ set of all rational numbers

$R \rightarrow$ set of all real numbers

$W \rightarrow$ set of all whole numbers $\{0, 1, 2, 3, \dots\}$

Cardinality of a Set

Cardinality of a set is the number of elements present in the set.

$$A = \{1, 2, 3, 4\}$$

$$|A| = 4$$

$$\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\} \infty$$

$$|X| = |Y|$$

X and Y have same cardinality.

$$|X| \leq |Y|$$

no. of X and Y have less elements less than or equal to no. of elements in Y.

Q. $|X| \leq |Y|$ & $|Y| \leq |X|$

X and Y have equal no. of elements

SUBSET:- If ^(all) any elements of set A is also element of set B then A is a subset of B.

$$B = \{1, 2, 3, 4\}$$

$$A = \{2, 3\}$$

$$A \subseteq B$$

Every set is a subset of itself.

hence it is called a trivial set.

Note:- ϕ

Proper subset not a true subset of

$$A = \{1, 2\}$$

$$A' = \{2, 3\}$$

$$A'' = \{1, 2\}$$

$$A''' = \{1, 2, 3\}$$

ϕ अपने म इसका

SET $\rightarrow$

Some of the singleton and disjoint

(1) Finite:-

no. of elements

E.g:-

$$S = \{x : x \in \mathbb{R}\}$$

(2) Infinite:-

$$S = \{x : x \in \mathbb{R}\}$$

(3) Empty:- $\{x : x \in \mathbb{R}\}$

(4) Singleton set

$$S = \{x : x \in \mathbb{R}\}$$

ness of

$$2, 3, \dots \} \infty$$

$$A' = \{2, 3\}$$

$$A'' = \{1, 4\}$$

$$A''' = \{3, 4\}$$

Proper subset of A

① अपने आप में terminal subset है नही इसका कोई proper subset नही होता है।

ality

to less than
cents in %.

SET $\rightarrow$ Classified into many types
Some of them are finite, infinite, empty,
singleton, equal, equivalent, overlapping
and disjoint set.

of elements

of set A
then A is a

① Finite :- A set which contains a definite no. of elements is called a finite set.

Ex. 9: -

E.g.:-
 $S = \{x : x \in \mathbb{N} \text{ and } 100 < x < 500\}$

(2) Infinite :- infinite number of elements
 $S = \{x : x \in \mathbb{N}\}$

$$S = \{x : x \in N\}$$

③ Empty: $\{x: x \in \mathbb{N} \text{ and } 7 < x < 8\} = \emptyset$

empty :- $\{x : x \in \mathbb{N} \text{ and } 7 < x < 8\} = \emptyset$

④ Singleton set:- contains only one element

$$S = \{x : x \in \mathbb{N} \text{ and } 7 < x < 9\} = \{8\}$$

itself

Trivial set

⑤ Equivalent:- containing same no. of elements. (cardinality is same)

⑥ Equal:- containing same elements.
 $S = \{1, 2, 4\}$
 $S' = \{6, 2, 13\}$
 $S = S'$

⑦ Overlapping:- Two sets having at least one common element is called overlapping set

not overlapping $S_1 = \{1, 6, 2\}$ overlapping $S_2 = \{6, 3, 1\}$
 $S_4 = \{16, 23, 14\}$

⑧ Disjoint sets:- If two sets do not have any one element in common.

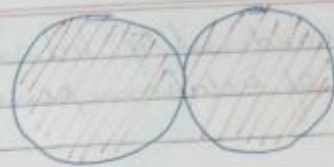
$S_2 = \{6, 3, 1\}$
 $S_4 = \{16, 23, 14\}$

SET OPERATION

$U, \cap, -, \cup$ Cartesian product
 complement of a set

A B
 $A \cup B$

Venn diagram $\rightarrow$ graphical representation of set



20/9/22

Classmate

Complement Set
 $A - B$ A'

$A = \{1, 2, 3\}$
 $B = \{3, 5\}$

$A - B = \{1, 2\}$
 $B - A = \{5\}$

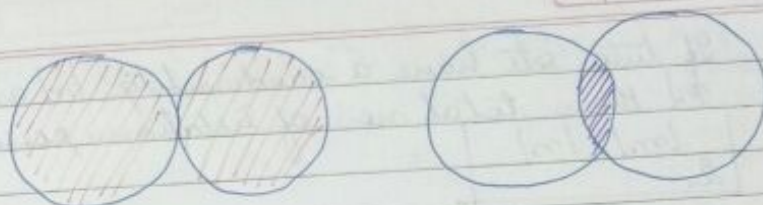
$A = \{x: x \in N\}$
 $A' = \{1, \dots, 29\}$

$A = \{a, 1\}$
 $B = \{b, 2\}$

$A \times B = \{(a, b), (1, 2)\}$

Let A and B are relation from

$A \times B \neq B \times A$
 $\downarrow \quad \downarrow$
 m elements n elements
 $= m \times n$ no. of elements
 $n \times n = n^2$



80/9/22

Classwork

Complement
 $A - B$

Set diff
 A'

$C \times P$ (Cartesian Product)

$$A = \{1, 2, 3\}$$

$$B = \{3, 5\}$$

$$A - B = \{1, 2\}$$

$$B - A = \{5\}$$

$$A - B \neq B - A$$

↑
elements only present in A

$$A = \{x : x \in \mathbb{N} \text{ and } x > 30\}$$

$$A' = \{1, \dots, 29\}$$

$$A = \{a, 1\}$$

$$B = \{b, 2\}$$

$$B \times A = \{\{b, a\}, \{b, 1\}, \{2, a\}, \{2, 1\}\}$$

$$A \times B = \{\{a, b\}, \{a, 2\}, \{1, b\}, \{1, 2\}\}$$

Let A and B are two sets then every relation from A to B is subset of $A \times B$.

$$A \times B \neq B \times A$$

$$\begin{matrix} \downarrow & \downarrow \\ m & n \text{ elements} \\ \text{elements} & \end{matrix}$$

$$= m \times n \text{ no. of elements}$$

$$n \times n = n^2$$

If two sets have a cardinality m and n then total no. of relations possible $2^{m \times n}$

$$R_1 = \{(a, b), (a, 2)\}$$

$$R_2 = \{(a, b), (1, b)\}$$

$$2^{12 \times 12} = 16$$

possible outcomes

A relation R on set A is said to be reflexive if $(x, x) \in R \forall x \in A$

if every element of the set is related to itself then it is a reflexive relation.

for e.g. - $A = \{1, 2, 3\}$ $A \neq \{1, 2, 3\}$

$$A \times A = \{(1, 1) (1, 2) (1, 3) (2, 1) (2, 2) (2, 3) (3, 1) (3, 2) (3, 3)\}$$

$$\begin{matrix} (1, 1) & (1, 2) & (1, 3) \\ (2, 1) & (2, 2) & (2, 3) \\ (3, 1) & (3, 2) & (3, 3) \end{matrix}$$

diagonal elements

$$2^{13 \times 13} = 512$$

(possible outcomes in relation)

$$R_1 = \{(1, 1), (1, 2)\} \dots$$

$$R_f = \{(1, 1) (2, 2) (3, 3)\}$$

reflexive for every element in A again $(1, 2)$ element, $(2, 1)$ not reflexive

31/12 Rf

Q $A = \{1, 2\}$

Reflexive $R_1 = \{(1, 1)\}$

$R_2 = \{(2, 2)\}$

$R_3 = \{(1, 1), (2, 2)\}$

21/09/22

Q $R_1 = \{(1, 1)\}$

$R_2 = \{(2, 2)\}$

Check a set

$$R_1 = \{(1, 1)\}$$

$$R_2 = \{(2, 2)\}$$

$|m| \rightarrow 2$

$|m| \times |m|$

Q Relation

How many on set A ?

3/11/2 $R_f = \{(1,1), (2,2), (4,4), (3,3), (4,3)\}$

↓
extra 0
chis 0 n set
the 1/1 reflexive
given at
elements set
A for part
2/

Q $A = \{1, 2, 3\}$

Reflexive $R_1 = \{(1,1), (2,2), (3,3)\}$

$R_2 = \{(1,1), (2,2), (3,3)\}$

$R_3 = \{(1,1), (2,2), (3,3), (2,3)\}$

21/09/23

Classwork

Q $R_1 = \{(x,y) \mid x-y \text{ is an integer}\}$

$R_2 = \{(x,y) \mid x-y \text{ is an odd integer}\}$

Check which one of the reflexive set is a set of all natural numbers.

$R_1 \begin{cases} \rightarrow (1,1) \rightarrow 0 \quad (1-1) \checkmark \\ \rightarrow (2,2) \rightarrow 0 \quad (2-2) \checkmark \\ \rightarrow (3,3) \rightarrow 0 \quad (3-3) \checkmark \end{cases}$

R_1 reflexive

$R_2 = \{(x,y) \mid x-y \text{ is a whole number}\}$

↓
Reflexive

$|m| \rightarrow 2^m$ (power set)

$|m| \times |m| \rightarrow 2^{|m| \times |m|}$ (number of possible relations)

Q Relation subset = $A \times B$

How many reflexive relations are possible on set A?

ity and
ions possible

iff

es

said to be
+ $x \in A$

of the
itself
xiii

$\{1, 2, 3\}$
 $(1,3) (2,1)$
 $(3,1) (3,2)$

ual elements

outcomes
relation

ment int.

ent, $\notin \{4,4\}$
it not reflexive

no. of possible reflexive relations

$A = \{1, 2\}$
 $A \times A = \{(1,1), (1,2), (2,1), (2,2)\}$

$2^{2 \times 2} = 2^4 = 16$

elements in cross product. possible no. of outcomes

$A \times A = \{(1,1), (1,2), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

$= 64 \leftarrow$ no. of possible reflex relations.

$|n| \times |n| = n^2$

$n^2 - n =$ Non diagonal elements

Diagonal elements

$2^{n^2 - n} =$ no. of possible reflexive relations

$2^{n^2} =$ no. of possible relations

Total number of non reflexive relations
 $= 2^{n^2} - 2^{n^2 - n}$

Q Can you tell the size of smallest no. of possible reflexive relations?

n is size

n^2 is largest possible

is relation if $x \in x$

$A = \{1, 2, 3\}$
 $A \times A = \{(1,1), (2,1), (3,1), \dots\}$

$R_1 = \{(1,2)\}$
 $R_2 = \{(1,2)\}$

Not even a pair of diagonal element should there.

Q Not reflexive relation

Q Smallest reflexive

largest reflexive (size)

Q $<$ is reflexive

$<=$ is not

$<=$ REF

no. of possible reflexive relation

$x = 4$

1, (2,2)

$= 16$

possible no. of outcomes

$2 \times 1 \times 1$

(2,1) (2,2)

(2,2) (3,3)

2×1

possible reflex relations

diagonal elements

is reflexive

om.

reflexive relations

smallest

m?

x

smallest

m?

x

smallest

m?

x

smallest

m?

x

smallest

m?

IRREFLEXIVE

A relation R on set A is said to be irreflexive if $x \not R x$ { x is not related to x } $\forall x \in A$, i.e. $(x, x) \notin R$.

diagonal elements नहीं होंगे।

$A = \{1, 2, 3\}$

$A \times A = \{(1,1) (1,2) (1,3) (2,1) (2,2) (2,3) (3,1) (3,2) (3,3)\}$

$R_1 = \{(1,2) (1,3)\} \rightarrow$ Irreflexive ✓

$R_2 = \{(1,2) (2,2) (1,3)\} \rightarrow$ ना है Irreflexive

ना है reflexive

cause (2,2) आया

तो irreflexive नहीं है

अब reflexive के लिए

(1,1) (3,3) नहीं है।

Not even a single pair of diagonal element should be there.

Q Not reflexive relation = reflexive? FALSE (Above e.g)

Q Smallest irreflexive set = $\{\}$ empty set.

largest irreflexive set = $n^2 - n$ (size)

Q $<$ is irreflexive on set of natural numbers TRUE

$\leq$ is not irreflexive on set of natural numbers. TRUE

$\leq$ REFLEXIVE है जायगा।

Symmetric Relation

A relation R on set A is said to be symmetric if $x R y$ then $y R x \forall (x, y) \in A \times A$

$$A = \{1, 2, 3\}$$

$$A \times A = \{(1,1) (1,2) (1,3) (2,1) (2,2) (2,3) (3,1) (3,2) (3,3)\}$$

$$R_1 = \{(1,2) (2,1)\} \text{ Symmetric Relation.}$$

$$R_2 = \{(1,2) (2,1) (1,1)\} \text{ Symmetric relation.}$$

22/09/22

Not reflexive or irreflexive

Classmate (AVC)

$$R_2 = \{ \} \text{ symmetric set } \checkmark$$

reflexive $\times$
 irreflexive $\checkmark$

1	2	3
4	5	6
7	8	9
10	11	12

$$R = \{(2,3) (3,2) (1,1)\}$$

$S \checkmark \quad R \times \quad I \times$

$$R_2 = \{(1,2) (1,3) (3,1)\}$$

$S \times \quad R \times \quad I \times$

said to be
 $\forall x \forall y (x R y \rightarrow \exists z (x, y) \in A)$

0
 0}

ation.

symmetric
 relation.

line

✓

(3,2) (1,1)
 $\times \nexists x$

Q There is a relation $\perp^r$ set of all lines.
 Verify its a symmetric relation or not.

$$L_1 \perp^r L_2$$

Q X is a brother of Y and X and Y $\in$ all
 human beings. Verify that this relation
 is a symmetric relation.

False
 maybe Y sister $\exists$ 1

$$A = \{1, 2\}$$

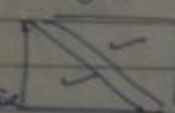
ANTI SYMMETRIC RELATION

$$\begin{matrix} \times & \times & \times \\ \neq & \neq & \neq \\ \neq & \neq & \neq \end{matrix} \begin{matrix} (1,1) & (1,2) \\ (2,1) & (2,2) \end{matrix}$$

A relation R. If $x R y$ and $y R x$ then
 $x = y \forall x, y \in A$, i.e. if $(x, y) \in R$ then
 $(y, x) \in R$. Only if $(x, y) \neq (x, y) \in A$.

$$R = \{ (1,2), (2,2) \}$$

$$R = \begin{matrix} 1 & 2 \\ 1 & 0 \\ 2 & 1 \end{matrix}$$



Auto
 Symmetric

matrix not $\times$
 all elements
 occur the same

non reflexive can be
 both anti-symmetric and
 symmetric. (only diag elements)

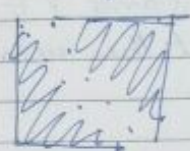
$$R_3 = \{ (1,1) (2,2) (1,3) \}$$

anti-symmetric $\rightarrow$ True
 Symmetric $\rightarrow$ False.

on
 $\sum$ all used are antisymmetric
 $\exists \neq 1$

ASYMMETRIC RELATION

A relation R is said to be asymmetric if xRy and $y \not R x$ then $x \neq y \forall (x,y) \in R$
 i.e. if $(x,y) \in R$ then $(y,x) \notin R \forall (x,y) \in R$



$$A = \{1, 2, 3\}$$

$$R_1 = \{(1,2), (1,3), (1,1)\} \times$$

not asymmetric
 antisymmetric ✓

$$R_2 = \{(1,2), (1,3), (2,3)\} \quad S \times$$

Anti S ✓
 AS ✓

Q. Verify every asymmetric is antisymmetric.
 TRUE
 self loop या diagonal elements
 हटा दिजे वम।

Every antisymmetric is asymmetric
 → False

Antisymmetric = asymmetric + diagonal elements

TRANSITIVE RELATION:-

A relation R on a set A is said to be transitive if xRy and yRz
 then $(xRz) \forall x, y, z \in A$

i.e. if $(x,y) \in R$

$$A \rightarrow B$$

$$R_1 = \{ \}$$

$$R_2 = \{(1,2), (2,3)\}$$

$$R_3 = \{(1,2), (2,3), (1,3)\}$$

$$R_4 = \{(1,2), (2,3), (1,3), (1,1), (2,2), (3,3)\}$$

$$R_5 = \{(1,2), (2,3), (1,3), (1,1), (2,2), (3,3), (1,2), (2,3), (1,3)\}$$

$$\textcircled{R} \leq \textcircled{S}$$

$$x \leq y$$

ITION

to be asymmetric
if then $x=y \vee (x,y) \in R$

$(y,x) \notin R \vee (x,y) \in R$

$\{1,2\}$

$\{(1,2), (1,1)\} \times$

not asymmetric

symmetric ✓

$\{2,3\}$ SX

antis ✓

AS ✓

9

trans is antisymmetric
TRUE

diagonal elements

331

asymmetric
→ False

symmetric + diagonal
elements

A is said to be
and $y R z$
 $x, y, z \in A$

ie if $(x,y) \in R$ and $(y,z) \in R$ then $(x,z) \in R$.

$A \rightarrow B, B \rightarrow C$

$R_1 = \{ (1,1) (2,2) \}$
 $(x,y) (x,y) =$ नहीं है तो कोई बात नहीं

$R_2 = \{ (1,2) (2,3) \}$ Transitive नहीं है
 $(x,y) (y,z)$
 $(x,z) \rightarrow$

$R_3 = \{ \}$ Transitive ✓

$R_3 = \{ \}$ IRREFLEXIVE ✓, SYMMETRIC ✓

$R_4 = \{ (1,1) (1,2) (2,1) \}$ transitive

$(1,1) (1,2) (1,2) (2,1)$
 $(1,1) (1,2) (1,2) (1,1) (2,2) (2,1)$

$\textcircled{1} \leq \textcircled{2}$

$\textcircled{2} \leq \textcircled{1}$

$\textcircled{1} \leq \textcircled{2}$

नहीं है तो
transitive नहीं है

$x \leq y$

$y \leq x$

$x \leq z$

$A = \{1,2\}$

$A \times A = \{ (1,1) (1,2) (2,1) (2,2) \}$

$(1,2)$
 $(2,1)$
 $(2,2)$

$1 \leq 1$

$1 \leq 2$

$2 \leq 1$

$1 \leq 2$

27/9/22

Classwork

Equivalence Relation

A relation R on a set A is said to be equivalence relation if it is reflexive, symmetric, transitive.

$$A = \{1, 2, 3\}$$

$$R = \{(1,1), (2,2), (3,3)\}$$

Ref ✓

Sym ✓

Transitive ✓

E.g of equivalence

$$A = \{1, 2, 3\}$$

$$R = \{(1,1), (2,2), (3,3), (2,1), (1,2)\}$$

Ref ✓

Sym ✓

Transitive ✓

$$A = \{1\}$$

Ref X Equivalent X

• Cardinality of smallest relation

largest relation

how many relations

Transitive

$$\emptyset$$

$$n^2$$



Sym

$$\emptyset$$

$$\frac{n^2 - n}{2}$$

Ref Sym

$$\emptyset$$

$$\begin{aligned} * \text{ IMF} \\ n + n^2 - n \\ = \frac{n^2 - n}{2} \\ = \frac{n(n-1)}{2} \end{aligned}$$

Sym

$$\emptyset$$

$$n^2$$

Ref

Ref

$$\{1\}$$

$$n^2 - n$$

Reflexive

$$n$$

$$n^2$$

Cardinality of largest relation

largest relation

is said to
it is
transitive.

valence

$(2,1)(1,2)$

relation

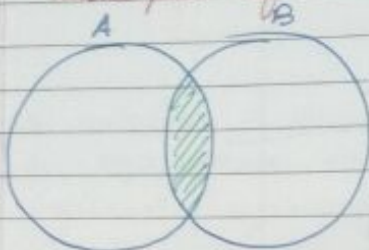
Cardinality of smallest relation	Reflexive	Symmetry	Transitive
n	$\{ \}$	ϕ	ϕ
largest relation	n^2	n^2	n^2
how many relation	2^{n^2-n}	$2^n \cdot 2^{\frac{n^2-n}{2}}$	$\times$
		$* \text{ L.H.F}$ $n + \frac{n^2-n}{2}$ $= \frac{2n + n^2 - n}{2}$ $= \frac{n(n+1)}{2}$ $2^n \cdot 2^{\frac{n(n+1)}{2}}$	$\frac{n^2-n}{2}$ $\frac{n^2-n}{2}$

For anti-symmetric

$$\begin{bmatrix} a & b & c & d & e & f & g & h & i \end{bmatrix}$$

$d=1$
 $b=2$
 $b=1$
 $d=2$

Principle of Inclusion / Exclusion



$$|A \cup B| = |A| + |B| - |A \cap B|$$

Sum of the numbers of elements of each set minus their intersection

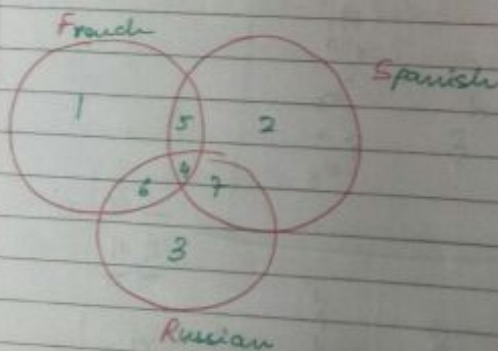
Q.

CS
(25)

Maths
(13)

CS + Maths
(3)

$$\begin{aligned} 25 + 13 - 3 \\ = 25 + 10 \\ = 35 \end{aligned} \leftarrow \text{total students}$$

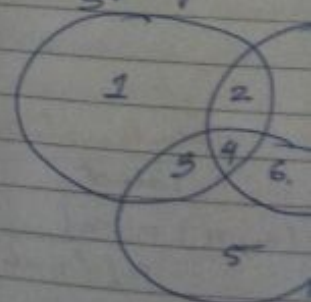


25/9/20

Q. There are 10 in JAVA, 15 programming those who know Python. Calculate

$$\begin{aligned} \text{Total} &= 10 + 15 \\ &= 20. \end{aligned}$$

Q. Suppose there are only one person who knows only 4 people can speak but 2 people Spanish, 3 can language



Pigeonhole Principle

If there are pigeon holes the one pigeonhole If $k+1$ or more boxes then

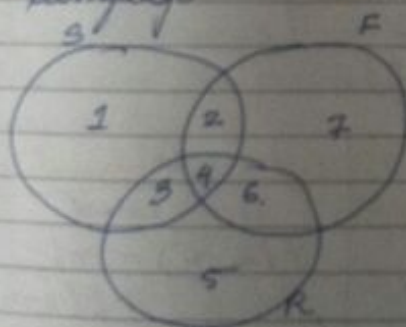
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Classwork

- Q There are 10 employees who love programming in JAVA, 15 employees are there who love programming in Python and 5 employees are there who love to program in Java Python. Calculate total.

$$\text{Total} = 10 + 15 - 5 \\ = 20.$$

- Q Suppose there is a seq of ~~one~~^{one} ~~set~~^{set} den in which only one person knows Spanish, ^{only 7} knew only French, 5 know Russian only 4 people can speak all three languages but 2 people can speak French & Spanish, 3 can speak Spanish & Russian language



$$1 + 7 + 5 + 4 - 2 - 3 - 6 \\ = 6$$

$$|A| + |B| + |C|$$

$$|A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| +$$

$$|A \cap B \cap C|$$

objects
total
n+2

Pigeonhole Principle

If there are more pigeons than pigeon holes then there must be at least one pigeonhole with ^{at least} 2 pigeons in it.
If $k+1$ or more objects are placed into the k boxes then there is at least one box

How many ways there are to select
1st prize, 2nd prize, 3rd prize ~~also~~
among 100 people

$$^{100}C_1 \cdot ^{99}C_1 \cdot ^{98}C_1$$

$$\boxed{100 \times 99 \times 98} \rightarrow \text{Possible ways}$$

How many permutation of the letters
A B C D E F G H are possible that contains
the string 'ABC'.

$$\begin{aligned} & \text{6! ways.} \\ & \frac{1 \times 1 \times 1 \times 2 \times 2 \times 2 \times 2 \times 2}{32 \text{ ways}} \quad \text{6! ways.} \quad \frac{1 \times 2^5 \times 5!}{32 \times 125} = 4000 \end{aligned}$$

$$\begin{aligned} 6! &= 24 \times 6 \\ &= 144 \end{aligned}$$

If there are 10 members and we
have to make a team in which 5
members per team are allowed then
possibilities

$$^{10}C_5 = \frac{10!}{5!5!}$$

Q Prove that $n^3 - n$ is divisible by 3 for $n \geq 1$

Base step:- $n^3 - n = P(n)$
 $\Rightarrow 1^3 - 1 = 0 \div 3 = 0$
 $P(1)$ is divisible by 3

Inductive step:-
 $P(k) = k^3 - k$ is divisible by 3

Verify:- $P(k+1) = (k+1)^3 - (k+1)$
 $= k^3 + 3k^2 + 3k + 1 - k - 1$
 $= \underline{(k^3 - k)} + \underline{3(k^2 + k)}$
 $\downarrow$
 we have proved that it is divisible by 3

$\therefore n^3 - n$ is divisible by 3.

Q Prove that $n^3 + 2n$ is also divisible by 3 for $n \geq 1$

Base step:- $P(n) = n^3 + 2n$
 $P(1) = 1^3 + 2(1) = 3$ Yes it is divisible by 3.

Inductive step:-
 $P(k) = k^3 + 2k$ is divisible by 3.

Verify:- $P(k+1) = (k+1)^3 + 2(k+1)$
 $= k^3 + 3k^2 + 3k + 1 + 2k + 2$
 $= \underline{(k^3 + 2k)} + \underline{(3k^2 + 3k + 3)}$
 $= \underline{(k^3 + 2k)} + \underline{3(k^2 + k + 1)}$
 $\downarrow$
 we have proved it divisible by 3

$\therefore n^3 + 2n$ is a

Q $1 + 2 + 2^2 + \dots$

Base step:- $P(0)$
 Inductive step:-
 $1 + 2 + 2^2$

Verify:- $P(k+1)$
 $= 1 + 2$

$\downarrow$
 $= \frac{k+1}{2} \cdot \frac{k+1}{2}$
 $= \frac{(k+1)^2}{4}$
 $= \frac{k^2 + 2k + 1}{4}$
 $= \frac{k^2}{4} + \frac{k}{2} + \frac{1}{4}$

Hence proved

Complete Induction
 we check base case
 consecutive steps
 assume the P(n) is true
 verifies it for P(n+1)

Mathematical induction
 in induction, we assume or not a statement is true. After that we verify P(n+1).

$\therefore n^3 + 2n$ is divisible by 3 [Hence proved]

Q. $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all non negative integers, $n \geq 0$

Base step:- $P(0) = 2^{0+1} - 1 = 1$

Inductive step:-

$$1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1 \text{ is } \underline{\underline{\text{also true}}}$$

Verify:- $P(k+1)$

$$= 1 + 2 + 2^2 + \dots + 2^k + 2^{k+1} = 2^{k+2} - 1$$

$$\begin{aligned} & \downarrow \\ & \equiv 2^{k+1} \cdot 2 - 1 \\ & \equiv 2^{k+1} + 2^{k+1} \\ & = 2 \cdot 2^{k+1} - 1 \\ & = 2^{k+2} - 1 \quad (\text{LHS} = \text{RHS}) \end{aligned}$$

Hence proved.

Complete Induction:- In complete induction we check base step and two or three consecutive steps as well. Then we can assume the $P(k)$ also true and then verifies it for $P(k+1)$.

Mathematical induction:- In mathematical induction, we only check ^{the base step} whether it is true or not and then assume that $P(k)$ also true. After that, verifies it for $P(k+1)$.

Principle of Well Ordering:- Every non-empty subset of natural numbers has a smallest element. Thus, if 'S' be a non empty subset of natural numbers then there exists $p \in S$ such that $p \leq q \forall q \in S$.

7/10/22

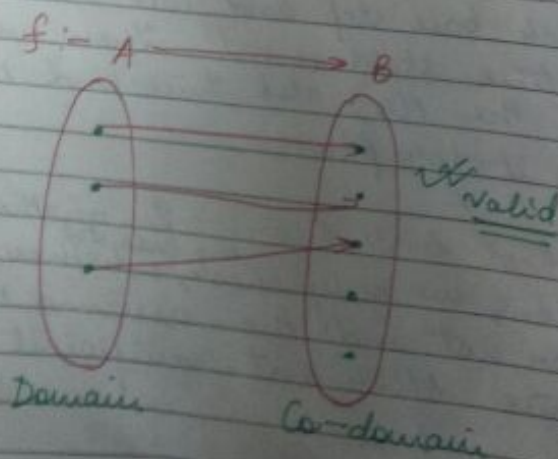
Classwork

Functions:-

A relation function f from a set A to set B is called function if each element in A we can assign unique element of B .

Conditions

- ① $a \in A$ uniquely mapping होनी चाहिए
- ② कोई भी $a \in A$ में value नहीं होनी चाहिए जिसका B में कोई unique value न हो।



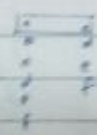
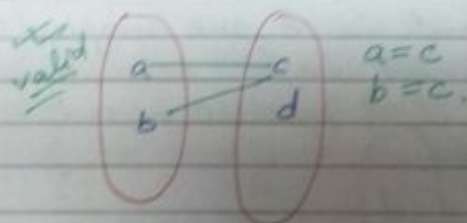
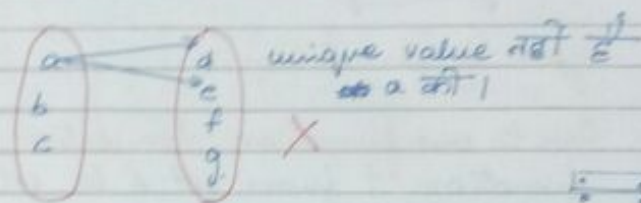
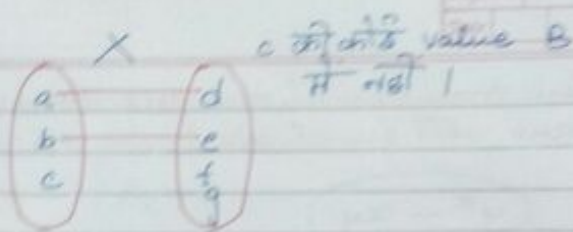
Range :-

Q. How many we have

every non-
natural
if '5'
natural
p. 5 such

a set A to
each
unique

not
not
unique



Range :- no. of possible outcomes for given input

$$R \subseteq \text{Codomain}$$

Q. How many functions are possible on set A and B.
 we have having an element $x \in Y$ is exp.



$$y \cdot y \cdot y \dots x \text{ times} = y^x$$

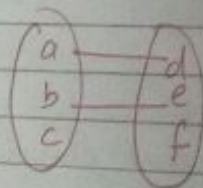
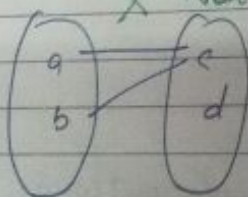
Q How many relations are there which are not functions?

$y^x - xy$

Types of functions:-

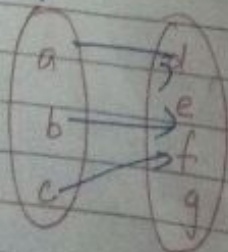
One to one function / injection:- A function f from set A to set B is said to be one-to-one if no two elements in A are mapped to same element in B .

X Not possible in one-to-one



X {NOT A FUNCTION}

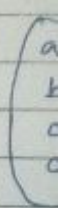
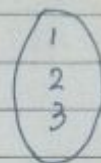
For one-to-one function:- No of element in set A and set B must satisfy $|X| \leq |Y|$



$$1 \cdot {}^y P_1 = {}^y P_1 \cdot {}^{y-1} P_1 = y \cdot (y-1) \cdot (y-2) \dots 1$$

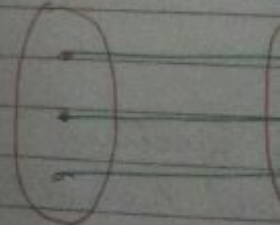
$$= y \cdot (y-1) \cdot (y-2) \dots 1$$

How many possible f/p



Onto function

A function f from set A to set B is said to be onto if each element in B has at least one pre-image in A .



Range of

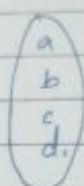
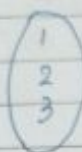
Q If there are more than one elements in B then there is no onto function.

If function is one-to-one and onto, it is a bijection.

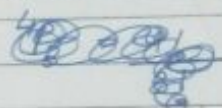


How many one to one functions are possible from A to B?

4P_3

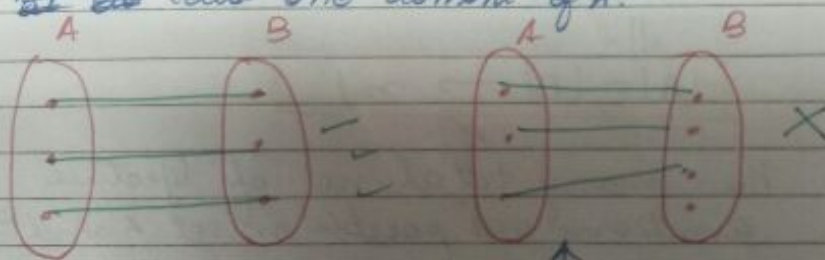


$$4 \times 3 \times 2 = 24$$



Onto function :-

A function f from set A to set B is onto if each element of B is mapped to at least one element of A.

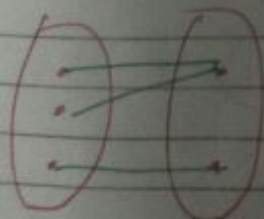


Range of $f = B$

For example

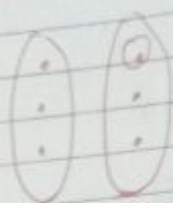
Q If there is a function which is one-to-one then is it onto as well?
 No

If function is onto then it is always one-to-one as well. False



Onto $\frac{19}{8}$
 42 one-to-one
 नहीं 1

Q. How many onto functions are possible if $|A| = |B|$



$$n \times (n-1) \times (n-2) \times \dots \times 1 = n!$$

Bijection function:- A function $f: A \rightarrow B$ is said to be bijective if function is both one to one and onto.

$|A| \leq |B| \rightarrow$ one to one

$|A| \geq |B| \rightarrow$ onto

$$\Downarrow$$

$$|A| = |B| \rightarrow n!$$

How many total no. of bijective functions are possible on set A and B?

Permutation function:- A permutation of a function is one to one and onto both.

$$A = \{1, 2, 3, 4, 5\}$$

$$P: A \rightarrow A$$

$$P(1) \rightarrow 4$$

$$P(4) \rightarrow 3$$

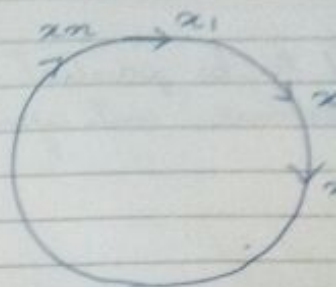
$$P(2) \rightarrow 2$$

$$P(5) \rightarrow 1$$

$$P(3) \rightarrow 5$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}$$

Cycling Permutation

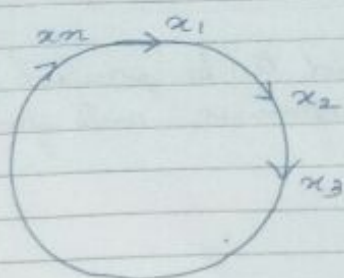


$$A = \{1, 2, 3, 4\}$$

$$(2 \ 1 \ 3 \ 4)$$

$$(5, 4, 2) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Cycling Permutation



$$P(x_1) = x_2$$

$$P(x_2) = x_3$$

$$P(x_n) = x_1$$

$$A = \{1, 2, 3, 4, 5\}$$

$$(2 \ 1 \ 3 \ 4) \text{ @}$$

$$(2 \ 1 \ 3 \ 4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 1 & 4 & 2 & 5 \end{pmatrix}$$

2 के बाद आया 1 तो 2 के नीचे 1
1 के बाद 3 तो 1 के नीचे 3

$$(5, 4, 2) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 3 & 2 & 4 \end{pmatrix}$$

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Classwork

Partial Order Relation

A relation R is said to be partial order relation if R is reflexive, anti symmetric and transitive.

R
 S
 T } $\rightarrow$ Equivalence.

Partially Ordered Set:- A set A which satisfies the condition of partial ordering relation is called partial ordered set.

$A \rightarrow$ POR.

$[A; R]$

$A = \{1, 2, 3\}$

$R_1 = \{(1,1) (2,2) (3,3)\}$

$R_1 \rightarrow$ POR $\checkmark$

$R_2 = \{(1,1) (2,2) (3,3) (1,2) (2,3) (1,3)\}$

If A is any set of real number then $[A; \leq]$ is a POSET

$$1 \leq 1 \checkmark$$

$$2 \leq 3 \checkmark$$

$$2 \leq 3 \checkmark$$

Totally Ordered (TOS)

A poset $[A; R]$ is a totally ordered set if the elements are comparable.

Again $a R b$
 $b R a$ $\checkmark$

Every TOS is a POSET.



If A is any set of real number then $[A; \leq]$ is a POSET

Ref $\checkmark$ $1 \leq 2 \checkmark$
Anti $\checkmark$ $2 \leq 4, 4 \leq 2$
Trans $\checkmark$ $1 \leq 2, 2 \leq 4 \Rightarrow 1 \leq 4$

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Classmate

Page No.	1	2	3
Date			

Partial Order Relation

A relation R is said to be partial order relation if R is reflexive, anti symmetric and transitive.

R
 S
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Partially Ordered Set:- A set A which satisfies the condition of partial ordering relation is called partial ordered set.

$A \rightarrow$ POR.
 $[A; R]$

$A = \{1, 2, 3\}$
 $R_1 = \{(1,1) (2,2) (3,3)\}$
 $R_1 \rightarrow$ POR $\checkmark$

$R_2 = \{(1,1) (2,2) (3,3) (1,2) (2,3) (1,3)\}$

If A is any set of real numbers then $[A; \leq]$ is a POSET

$1 \leq 1 \checkmark$ Reflexive

$2 \leq 3$ $3 \leq 4$

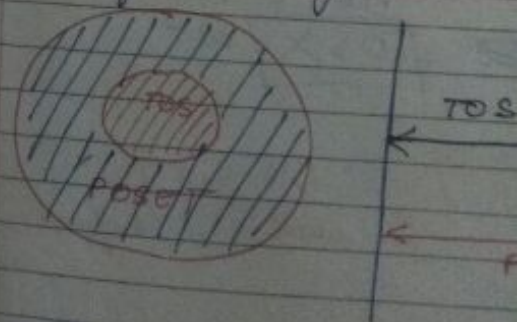
$2 \leq 3 \checkmark$ Antisym

Totally Ordered Set:-
(TOS)

A poset $[A; R]$ under a totally ordered set the element in A

Again $a R b$ or $b R a$ $\forall a, b$ exist in A

Every TOS is POSET
may or may not be



If A is any set of real numbers prove $[A; \leq]$ is

Ref $\checkmark$ $1 \leq 1, 2 \leq 2, 3 \leq 3$
Anti $\checkmark$ $2 \leq 4, 4 \leq 2 \times$ not
Trans $\checkmark$ $1 \leq 2, 2 \leq 3, 1 \leq 3$

ation

be partial order
ie, antisymmetric

set A which
of partial
led partial

$(1,2), (2,3), (1,3)$

number
POSET

$$1 \leq 1 \checkmark \text{ Reflexive}$$

$$2 \leq 3 \quad 3 \leq 4 \quad 2 \leq 4 \text{ Transitive } \checkmark$$

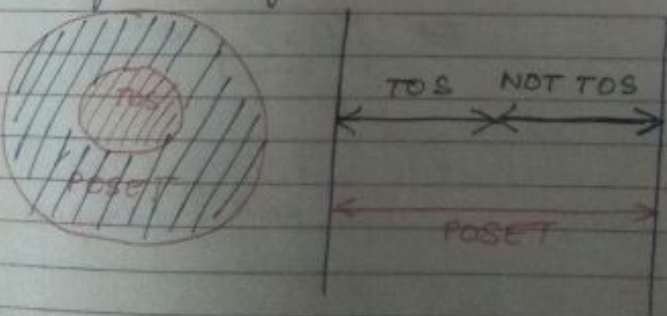
$$2 \leq 3 \checkmark \text{ Antisymmetric}$$

Totally Ordered Set:- or (literally ordered set)
(TOS)

A poset $[A; R]$ under relation R is called a totally ordered set, if every pair of the element ~~in~~ ⁱⁿ ~~are~~ ^{are} comparable.

Again $a R b$ or relation exist or not $\frac{y}{d}$ $\frac{y}{d}$
 $b R a$ ~~or~~ $\nexists$ exist or not $\frac{y}{d}$ $\frac{y}{d}$ $\forall a, b \in A$.

Every TOS is POSET. But a POSET may or may not be TOS.



If A is any set of positive integer then prove $[A; \div]$ is TOS.

Ref
pos
prop
anti

Ref $\checkmark$ $1 \div 1, 2 \div 2, 3 \div 3 \dots$

Anti $\checkmark$ $2 \div 4, 4 \div 2$ ~~or~~ $\nexists$ relation positive integer

Trans $\checkmark$ $1 \div 2, 2 \div 3, 1 \div 3$ ~~or~~ $\nexists$ relation positive integer

Alexa is comparable.

same features
or diff? if the integer
or diff? if decimal
if properties are diff.

POSET and suppose

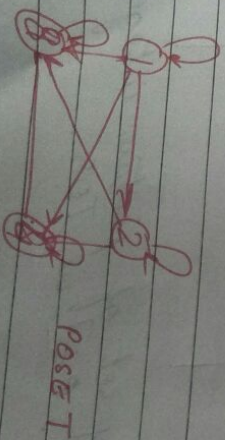
Q. Suppose there is a set $A = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$. Prove that power set of this set under the relation subset is TOS or not?
[P(A) $\supseteq$ C] is TOS or not?

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

subset
 $\{\emptyset\} \rightarrow \{a\}$
 $\{\emptyset\} \rightarrow \{b\}$
 $\{a\} \rightarrow \{a, b\}$
 $\{b\} \rightarrow \{a, b\}$
 if comparable
 or not?

TOS ✓

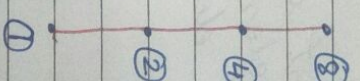
Q. $D_8 = [4, 1] = \{1, 2, 4, 8\}$
if poset is (surface).



POSET

Hasse Diagram

- ① Every element
- ② No upper element
- ③ No lower element
- ④ No direct



$A = \{1, 2, 3\}$

[P(A) $\supseteq$ C]

{1, 2, 3}

ϕ is minimal element

[{1, 2, 3}, ϕ]

POSET and

integer
decimal

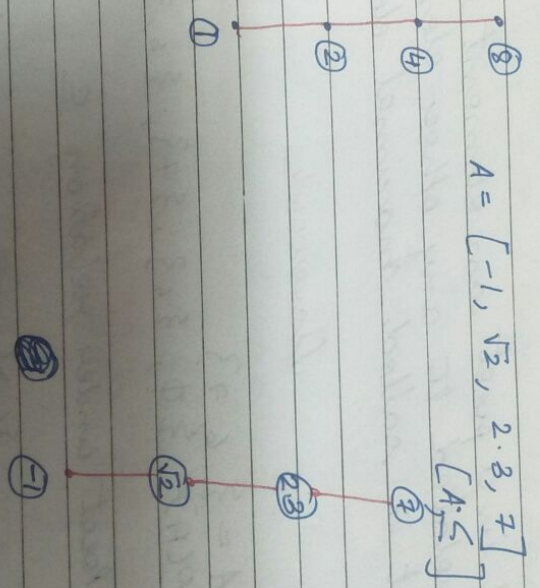
SET at suppose
b}, power set
under the

comparable

(suppose)

Hasse Diagram

- ① Every element of a set A is vertex.
- ② No reflexive loops are allowed.
- ③ No transitive edge should be mentioned.
- ④ No direction use make in hasse diagram.

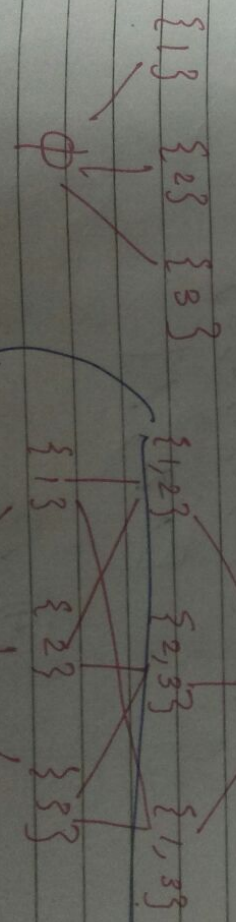


$$A = \{1, 2, 3\}$$

$\{1, 2, 3\}$ maximal element.

$$[P(A), \subseteq]$$

- $\{\phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$



ϕ is minimal element

$$[\{1, 2, 3, \phi\}]$$

$$[\phi, \{1, 2, 3\}]$$

subset of

comparable

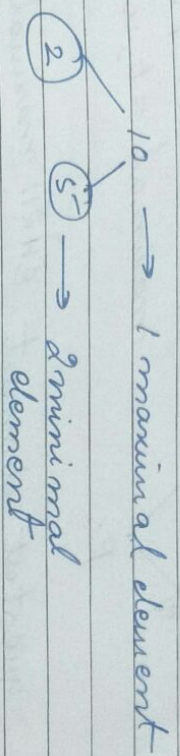
नहीं है साथ

if it same

level में, ये क्रमान-व Trees

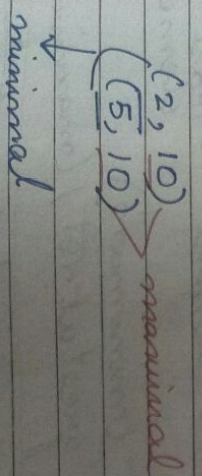
any poset can have more than one minimal element.

$[D, /]$
 $D = \{2, 5, 10\}$



any poset can have more than one minimal element also.

of minimal element दीना वी हमेशा LHS व भी होता होगा
 जबकि maximal हमेशा RHS व आता है।



Q Suppose given a Relation $R = \{(10, 0)\}$ find the maximal & minimal element.

यहाँ Poset पुनो बता दें $\rightarrow$ POR



यहाँ पर only ek element है compare करने के लिए 2 चीज चाहिए।

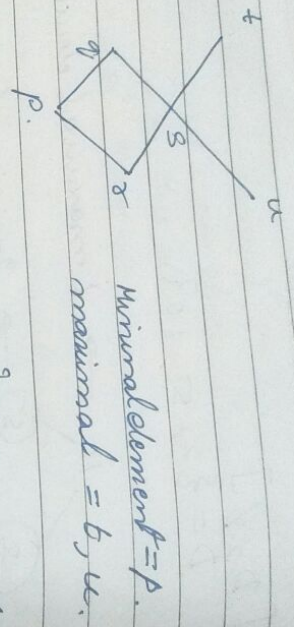
maximal $\rightarrow 0$
 minimal $\rightarrow 0$

Ref ✓
 Anti sym ✓
 Tr. ✓

आपस
 comparable नहीं
 कोई
 दोनो चीजें

11.

Greatest / least element in POSET

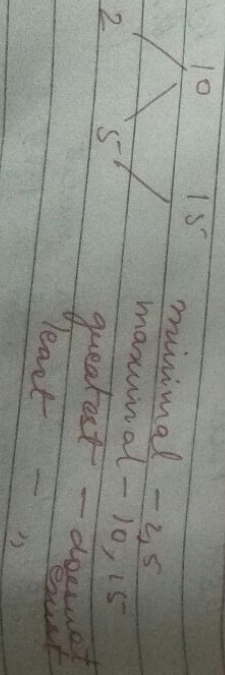


greatest element $\rightarrow$ $\exists! x \in A$ maximal
 least element $\rightarrow$ $\exists! x \in A$ minimal

single maximal element if exists
 is called greatest element
 single minimal element $\rightarrow$ least element

$P \rightarrow$ least (minimum)
 $\exists! x \in P$ greatest $\rightarrow$ (maximum)

Maximal $\exists!$ minimal element $\exists!$
 but least $\exists!$ greatest
 exist $\exists!$ simultaneously $\exists!$



Upper Bound and the

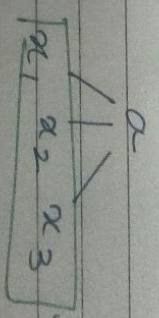
$[A; R]$

Upper Bound

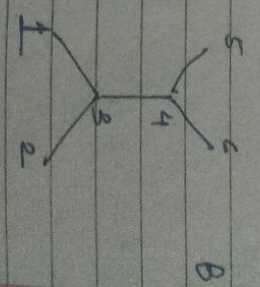
$a \in A$ is called U
 $(B \subseteq A)$ if $x \in R$

Lower Bound

$a \in A$ is lower bo
 $\exists! a \in R$ $\exists! a \in$
Upper Bound
 $\exists! a \in R$



$A = \{1, 2, 3, 4\}$



$UB \{1, 2, 3\}$

U.B
 $(3, R, 4)$ $(4, R, 5)$
 $(3, R, 5)$ $(4, R, 5)$
 $(3, R, 6)$
 $UB \{4, 5, 6\}$

it in POSET

element $= p$.
 $ad = b, u$.

maximal

minimal

but it exists

$t \rightarrow$ least element

m) (maximum)

element of
 the greatest
 only at

$-2, 5$
 $-10, 15$
 - distinct
 - subset

Upper Bound and Lower Bound

$[A; R]$

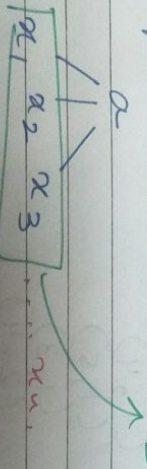
Upper Bound

$a \in A$ is called Upper Bound.
 $(B \subseteq A)$ if $x \leq_R a \forall x \in B$.

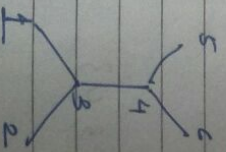
Lower Bound

$a \in A$ is lower bound of B , $B \subseteq A$
 if $a \leq_R x \forall x \in B$.

Upper Bound all lower bound subset of



$A = \{1, 2, 3, 4, 5, 6\}$



$B = \{3, 4\}$

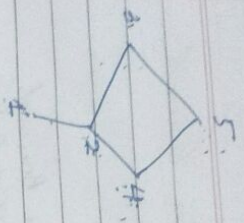
1 R 3	✓	all left
2 R 3	✓	front side
3 R 4	✓	lie and
1 R 4	✓	
2 R 4	✓	

$UB \{1, 2, 3\} = \{3, 4\}$

U.B
 $(3, R, 4)$ $(4, R, 5)$
 $(3, R, 5)$ $(4, R, 6)$
 $(3, R, 6)$

$UB \{4, 5, 6\} = (3, 4)$

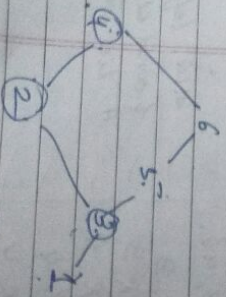
3 and 4 are not comparable if



UB $\{1, 2, 3\}$
 LB $\{4, 5\}$
 $\{5\}$

$\{1, 2, 3\}$
 LB $\{1, 3\}$
 UB $\{2, 4, 5, 3\}$

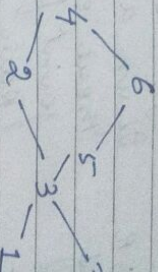
$\{1, 2\}$
 $\{1, 4\}$
 $\{1, 5\}$
 $\{1, 3\}$
 $\{2, 4\}$
 $\{2, 5\}$
 $\{2, 3\}$



$B = \{3, 4\}$

LB $\{1, 3\}$
 $\{2, 3\}$
 $\{3, 4\}$

UB $\{3, 5\}$
 $\{3, 6\}$
 $\{4, 6\}$
 $\{5, 6\}$
 $\{1, 3\}$

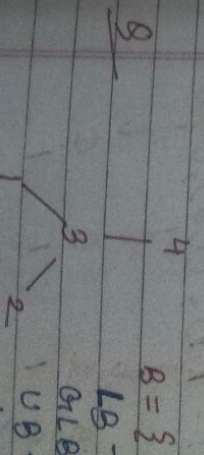


$B = \{3, 4\}$
 $LB = \{2\}$
 $UB = \{6\}$

Max of lower bound

Supremum $\rightarrow$ meet

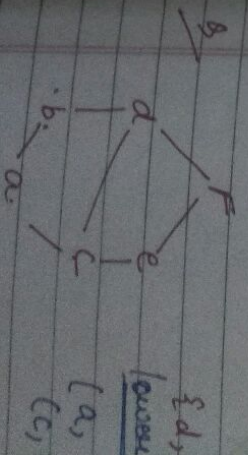
Minimum of upper bound
 $\rightarrow$ upper bound
 Infimum $\rightarrow$ join



$B = \{3\}$

LB $\{1\}$

UB $\{4\}$



LB $\{a\}$

UB $\{f\}$

4 अंकांचा प्रमाण्युत्तर

असंभव

{4,5}

{5,3}

{5,3}

{3,4}

UB {3,5}

{3,6}

{4,6}

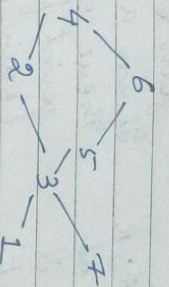
{5,6}

{6,3}

Teacher's Signature

Classwork

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$B = \{3, 4\}$

$B = \{3, 5\}$

$LB = \{2\}$

$LB = \{2, 1\}$

$UB = \{6\}$

$UB = \{5, 6\}$

Max of lower bound

→ greatest lower bound

Minimum → meet (V)

Greatest

lower bound

Minimum of upper bound

→ upper bounded → least upper bound.

Supremum → join (V)

UB

$B = \{1, 2\}$

$LB \rightarrow \phi$

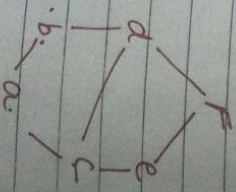
$ULB \rightarrow \phi$

$UB \rightarrow \{3, 4\}$

$ULB \rightarrow \{3, 3\}$



8



$\{d, e\}$

lower bound

(a, c)

(c, c)

(a, d)

(b, d)

(c, d)

upper bound

(d, f)

(d, d)

(c, f)

lower bound = $\{a, b\}$

upper bound = $\{f, f\}$

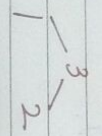
max = c.

min = d.

Teacher's Signature

Meet Semi lattice

If in a poset A, R for every pair of element there is a meet $(\wedge)$ exist then it is called meet semi lattice.



UB
 $\{2, 3\} \rightarrow 3$
 $\{1, 2\} \rightarrow 2$
 $\{1, 3\} \rightarrow 3$

LB

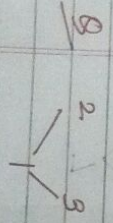
$\{1, 2\} \rightarrow \phi$
 $\{2, 3\} \rightarrow 2$
 $\{1, 3\} \rightarrow 1$

UB exists

GLB doesn't

So this structure is join semi

lattice but not a meet lattice.

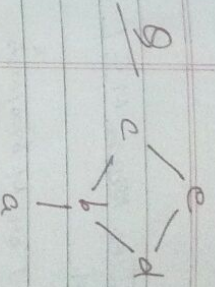


pairs :- $(1, 2), (1, 3), (2, 3)$

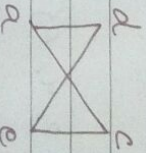
UB
 $1 \rightarrow 2$
 $2 \rightarrow 3$
 $3 \rightarrow \phi$
 $MSLV$
 $JSLX$

Join Semi lattice

If in a poset there exists a join for every pair of element then it is called join semi lattice.



JSL
 UB



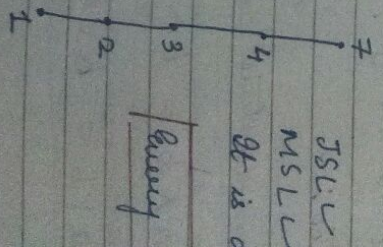
$JSLX$
 $MSLV$

lattice :- let $A: R$

lattice if for every is a meet as well

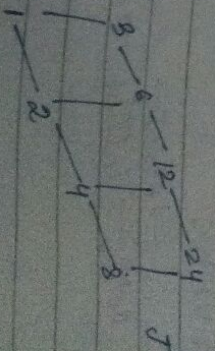
lattice = $JSL + M$

$A = \{1, 3, 4, 7\}$



$JSLV$
 $MSLV$
Every
 It is a

$SD = \{1, 2, 3, 4, 6, 8, 12, 24\}$



join semi lattice
in a poset
there exists a
point for every
pair of element
hence it is
called join
semi lattice

LB

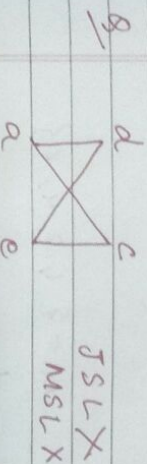
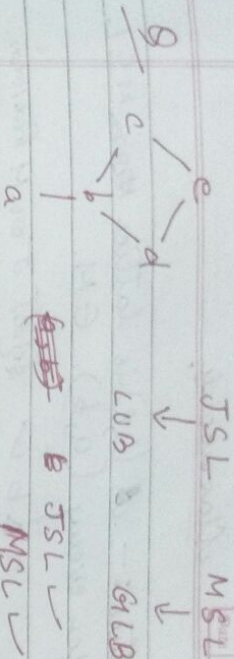
$\{1,2\} \rightarrow \emptyset$
 $\{2,3\} \rightarrow 2$
 $\{1,3\} \rightarrow 1$

GLB doesn't

is joint semi
lattice
it lattice

$(1,3), (2,3)$

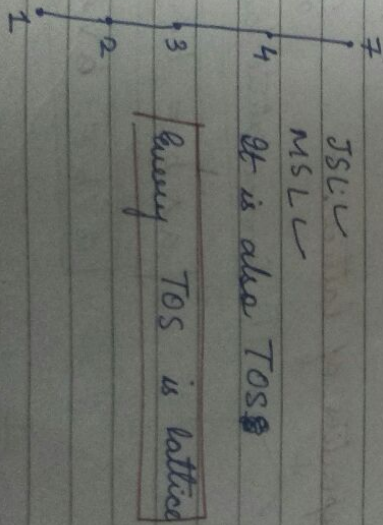
$\downarrow \downarrow$
3 ϕ
1 1



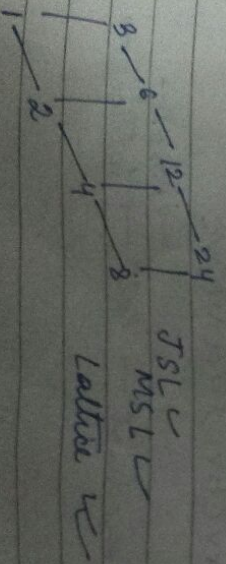
lattice :- let $A:R$ be a poset it is called
lattice if for every pair of elements there
is a mdtl as well as a joint

lattice = JSL + MSL

$A = [1, 3, 4, 7, \leq]$



$QD = \{1, 2, 3, 4, 6, 8, 12, 24\}$



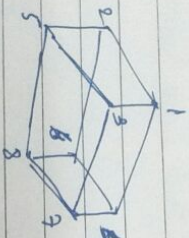
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Sublattice :- (L, R) is lattice ~~then~~ $M \subseteq L$ if for every $(a, b) \in M$.

$a \wedge b \in M$ $\checkmark$ $\exists$ $a \wedge b$ for M $\exists$ $a \wedge b$
 $a \vee b \in M$ $\checkmark$ $\exists$ $a \vee b$ for M $\exists$ $a \vee b$



$M = \{1, 2, 4, 6\}$



$M \subseteq L$
 $\checkmark$ $\exists$ $a \wedge b$ for M $\exists$ $a \wedge b$
 $\checkmark$ $\exists$ $a \vee b$ for M $\exists$ $a \vee b$

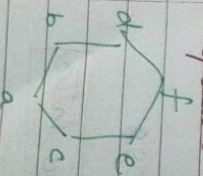
$M = \{5, 3, 7, 8\}$ $\checkmark$

$M = \{4, 6, 8, 9\}$ $\checkmark$

$M = \{1, 2, 3, 8\}$ $\times$

Properties of lattice :-

① Commutative law :-



$a \wedge b = b \wedge a$ $\checkmark$

$a \vee b = b \vee a$ $\checkmark$

② Associative law :-

$a \wedge (b \wedge c) = (a \wedge b) \wedge c$

$a \vee (b \vee c) = (a \vee b) \vee c$

Set $\rightarrow$ Cross product $\rightarrow$ Rel $\rightarrow$ Poset $\rightarrow$ Lattice $\rightarrow$ B4

③ Idempotent law

$a \wedge a = a$

$a \vee a = a$

④ Absorption law

$a \vee (a \wedge b) = a$

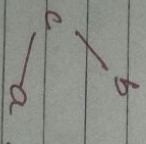
$a \wedge (a \vee b) = a$

⑤ Distributive law

$a \vee b = b$

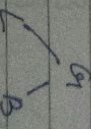
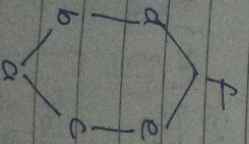
$a \wedge b = a$

a and b $\rightarrow$ each



⑥ Complete lattice :-

$[A, R] \rightarrow$ lattice
 Each non empty sub
 and join them it
 lattice.



Time ~~10:00~~ 11:01

For 14 questions
T total 211/157

2, 4, 6}

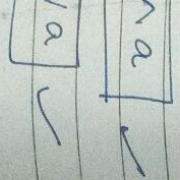
MSL

JSL

Sublattice

joint
meet

Law:-



three $\rightarrow$ 84

③ Idempotent law

$$a \vee a = a, \\ a \wedge a = a.$$

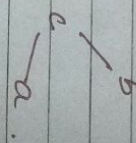
④ Absorption law

$$a \vee (a \wedge b) = a. \text{ same element.} \\ a \wedge (a \vee b) = a.$$

⑤ Distributive law (not applicable on all lattice).

$$a \vee b = b \\ a \wedge b = a.$$

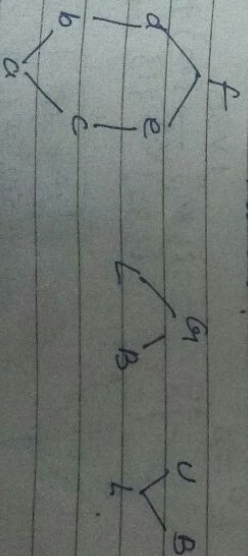
a and b ~~directly~~ connected to each other directly or indirectly.



⑥ Complete lattice:-

$[A, R] \rightarrow$ lattice

Each non empty subset of A has a meet and join then it is called complete lattice.

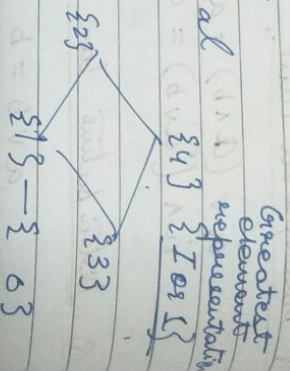
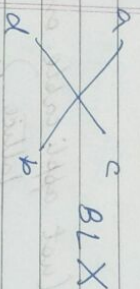


① **Bounded lattice**:- If a lattice has greatest and least ~~element~~ both it is called bounded lattice.

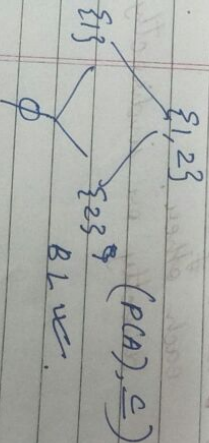
$$a, l \rightarrow le + ae.$$

$$le \rightarrow \text{minimal}$$

$$ae \rightarrow \text{maximal}$$



least element representation

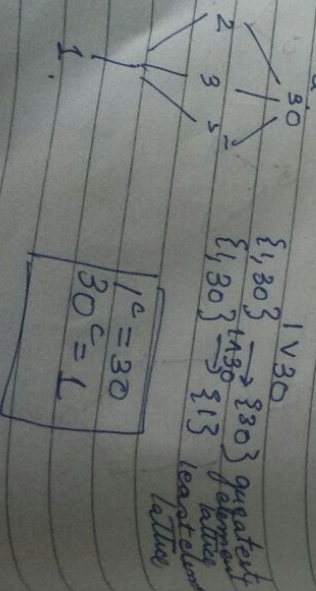


②

Complement any lattice:- If L is bounded lattice then $(a, b) \in L$ if $a \vee b \rightarrow$ Greatest $a \wedge b \rightarrow$ least element

$$a^c = b$$

$$b^c = a$$



$$\begin{matrix} 1^c = 30 \\ 30^c = 1 \end{matrix}$$

$$1, 2 = ?$$

$$1 \vee 2 = 2$$

$$1 \wedge 2 = 1$$

$$\begin{matrix} 1^c \neq 2 \\ 2^c \neq 1 \end{matrix}$$

$$(2, 3)$$

$$2 \vee 3 = 3$$

$$2 \wedge 3 = 1$$

$$\begin{matrix} 2^c = 3 \\ 3^c = 2 \end{matrix}$$

$$(2, 5)$$

$$2^c = 5$$

$$5^c = 2$$

For single element more than one

5 do complement
2 do
3 do

lattice has greatest
it is called

Greatest
element
representation
 $\{1\}$ $\{1, 0\}$

$\{3, 3\}$

$\{1, 3\} \rightarrow \{0, 3\}$

least element
representation

like
age list

ment

$1 \vee 3 = 3$
 $1 \wedge 3 = 1$
 $\{1, 3\} \rightarrow \{3, 0\}$ greatest
element
lattice
least element

$0^c = 3$
 $0^c = 1$

$1, 2 = ?$

$\{1, 2\} \vee 2 = 2$ $\rightarrow$ greatest element still
 $1 \wedge 2 = 1$ lattice is :-

$1^c \neq 2$
 $2^c \neq 1$

$(2, 3)$ $2 \vee 3 = 3$ (greatest)
 $2 \wedge 3 = 1$ (least)

$2^c = 3$
 $3^c = 2$

$(2, 5)$ $2^c = 5$
 $5^c = 2$

For single element, we can have
more than one element as complement

5 is complement of 2
2 is complement of 5
3 is complement of 5
5 is complement of 3

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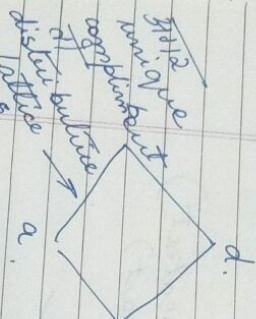
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Distributive lattice if

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$



$$a \vee (a) = a$$

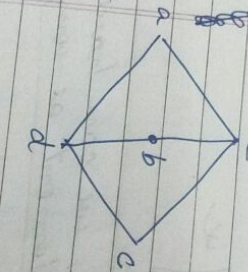
$$a \vee d = d$$

$$a \wedge d = a$$

$$b \wedge c = a$$

$$a^c = d$$

$$d^c = a$$



This lattice is not distributive lattice
 $a \wedge (e) = a$
 $d \vee (d) = d$

For the lattice to be distributive complement of element if exists must be unique.

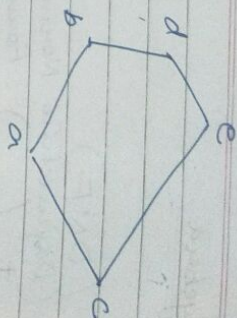
$$a \wedge c = d \text{ (lowest element)}$$

$$a \vee c = e \text{ (greatest)}$$

$$a^c = c$$

$$c^c = a$$

$a^c = [b, c]$
 Not distributive



$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

$$= a \vee a = a$$

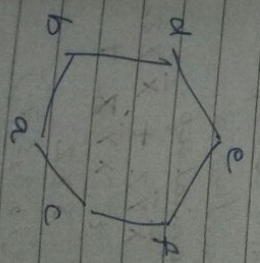
$$a \wedge c = a \text{ (least)}$$

$$c \wedge b = a$$

$$c \vee b = e$$

$$c \wedge d = e$$

$$c \vee d = a$$



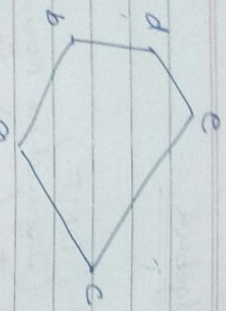
$$c^c = b$$

$\vee c$)

$$\begin{aligned} a \vee d &= d \\ a \wedge d &= a \\ a^c &= d \\ d^c &= a \end{aligned}$$

It is distributive

It is distributive
exists
(element
test ")



$$\begin{aligned} a \wedge (b \vee c) &= (a \wedge b) \vee (a \wedge c) \\ &= a \wedge e \\ &= a \end{aligned}$$

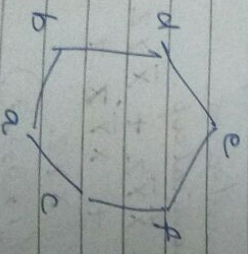
or

$$a \wedge c = a \text{ (least element)}$$

$$\begin{aligned} c \wedge b &= a \\ c \vee b &= e \end{aligned}$$

$$\begin{aligned} c \wedge d &= e \\ c \vee d &= a \end{aligned}$$

$$c^c = [b, d]$$



$$c^c = [b, d]$$

X Not distributive

Boolean Algebra

SOP (minterms) (DNF)
 (Disjunctive Normal Form)
 product

SOP doesn't need to contain all ~~minterms~~ literals.

Canonical SOP:- means it should contain all literals either in complemented or uncomplemented forms.

	X	Y	Z	f	functional output
0th	0	0	0	0	
1st	0	0	1	1	0 1 0 1
2nd	0	1	0	0	0 1 1 0
3rd	0	1	1	1	0 1 1 1
4th	1	0	0	0	1 0 0 0
5th	1	0	1	1	1 0 0 1
6th	1	1	0	0	1 1 0 0
7th	1	1	1	1	1 1 0 1

3rd 1 3rd 2nd
 corresponding minterms

$$f = \bar{x}\bar{y}z + \bar{x}yz + x\bar{y}z + xyz$$

$$\begin{aligned} &= \sum (1, 3, 5, 7) \\ &= \bar{x}(\bar{y}z + yz) + x(\bar{y}z + yz) \\ &= \bar{x}z(\bar{y} + y) + xz(\bar{y} + y) \\ &= \bar{x}z + xz \end{aligned}$$

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Duality Theorem:-

$$XOR \rightarrow \bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB$$

$$\begin{aligned} &= (\bar{A} + \bar{B}) \cdot (\bar{A} + B) \cdot (A + \bar{B}) \cdot (A + B) \\ &= \bar{A}\bar{A} + \bar{A}B + A\bar{B} + AB \\ &= \bar{A}B + A\bar{B} \end{aligned}$$

XNOR gate

$$\begin{aligned} &= AB\bar{C} + \bar{A}BC + ABC \\ &= (A + B + \bar{C}) \cdot (\bar{A} + B + C) \cdot (A + \bar{B} + C) \\ &= (A\bar{A} + A\bar{B} + AC + \bar{B}\bar{C} + \bar{A}B + \bar{A}C + \bar{B}C + \bar{C}C) \cdot (A + B + C) \end{aligned}$$

$$\begin{aligned} &= (A\bar{B} + B + C) \cdot (A + B + C) \\ &= A\bar{B} + B + C + A\bar{B}C + AB + AC + ABC + ABC \\ &= (A\bar{B} + C) \cdot (A + B + C) \end{aligned}$$

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(NFE)
Inclusive Normal
() Form

contain all

it should contain
complement

forms
output

IT 1 311 211
can corresponding
variables

initial set
 $\bar{X}\bar{Y}Z + \bar{X}YZ$

XYZ
(3,5,7)

$(\bar{X} + \bar{Y})$
 $(X + Y)$
 $(Y + \bar{X})$

$Z(Y + \bar{Y})$

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$\bar{X}Y + X\bar{Y}$
 $\bar{X}(\bar{X} + Y) + (X + \bar{Y})\bar{Y}$
 $\bar{X}\bar{X} + \bar{X}Y + X\bar{Y} + \bar{Y}\bar{Y}$

Duality Theorem:-

$OR \longleftrightarrow AND$
 $NOT \longleftrightarrow NOT$
 $XOR \longleftrightarrow X NOR$
 $NAND \longleftrightarrow NOR$

variables if
change sig

$XOR \rightarrow \bar{A}\bar{B} + \bar{A}B$

Dual

$(\bar{A} + \bar{B}) \cdot (\bar{A} + B)$
 $= \bar{A}\bar{A} + \bar{A}B + \bar{A}\bar{B} + B\bar{B}$
 $= \bar{A}B + \bar{A}\bar{B}$

XNOR gate

$ABC + \bar{A}BC + ABC$
 $= (A+B+C) \cdot (\bar{A} + B + C) \cdot (A + B + C)$
 $= (\bar{A}\bar{A} + \bar{A}B + \bar{A}C + B\bar{A} + B\bar{B} + B\bar{C} + C\bar{A} + C\bar{B} + C\bar{C}) (A+B+C)$

$= (\bar{A}\bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C} + ABC)$
 $= (\bar{B} + \bar{A}C + \bar{A}\bar{C} + B\bar{C}) (A+B+C)$
 $= (\bar{B} + \bar{A}C + \bar{A}\bar{C}) (A+B+C)$
 $= \bar{B} + B + \bar{B}C + AC + \bar{A}\bar{C} + AC$

$= B + \bar{B}C + AC + A\bar{B} + \bar{A}\bar{B}C + \bar{A}\bar{B}$
 $= B + \bar{B}C + AC + A\bar{B} + \bar{A}\bar{B}$
 $= (A\bar{B}C + \bar{A}\bar{B}C + ABC)$
 $= (A\bar{B}C) \cdot (A\bar{B}C) \cdot (A\bar{B}C)$

$$\begin{aligned} & \underline{2} \quad xy\bar{z} + \bar{x}y\bar{z} + \bar{y}z = 1 \\ & = \underline{\cancel{xyz + \bar{x}yz}} \quad \underline{z(xy + \bar{x}y)} + \bar{x}y\bar{z} = 1 \\ & = \underline{z(x + \bar{x})} + \bar{x}y\bar{z} = 1 \\ & = \underline{zx + z\bar{x}} + \bar{x}y\bar{z} = 1 \end{aligned}$$

$$= (x+y+z) \cdot (\overline{x+y+z}) \cdot (y+z)$$

~~$A\bar{A}C + A\bar{B}C + A\bar{A}C$~~

As function is said to be self dual if its dual is equivalent to itself

$$f(x, y, z) = (xy + yz + zx)$$

$$xy^2 + x^2 + 7x$$

$$-xy + xz + zy + xy$$

$$x^2 + 2x + 2y + 2y$$

$$f(x, y, z) = xy + yz + xz$$

Closework

- ① How many self dual functions are possible?
- ② Find the total number of expressions possible with n variables.
- ③ Find the total number of functions possible with n variables.

2

①

2

3

2 possible self dual

4 possible combinations

$$1 + \bar{x}y\bar{z} = 1$$

$$\bar{x} = 1$$

$$\bar{x}(y+z) = 1$$

be self dual
ent to itself.

$$zy + xy$$

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Assignment

$$f(x, y, z) = xy + yz + xz$$

- How many self dual functions with one variable
- Find the total number of boolean expression possible with one variable.
- Find the total number of self dual functions possible with one variable.

① A 2 possible combinations
0 1

② A f f f f f
0 0 1 1 1 0
1 0 0 1 1 1

4 possible boolean expression

③ f → 0, 1, A, $\bar{A}$
1 0 A $\bar{A}$ dual
x x

2 possible self dual functions.

2 variables
A B
0 0 4 possible combinations
0 1
1 0
1 1

③

A	B	f	f	f	f	f
0	0	0	0	1	0	0
0	1	0	0	1	0	0
1	0	0	1	0	0	0
1	1	0	1	0	0	0

$f \rightarrow 0, 1, A, B, \bar{A}, \bar{B}, \bar{A}\bar{B}, \bar{A}B, A\bar{B}, AB, A+B, \bar{A}+\bar{B}, AB+\bar{A}\bar{B}$

⑩ total outcomes.

③ ④ total outcomes

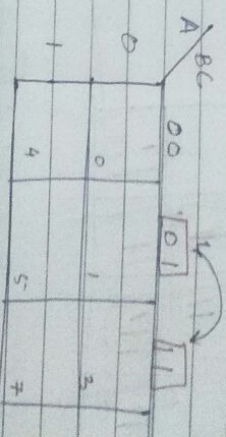
IMPORTANT

n variables
Total possible outcomes = 2^n
Total possible boolean expressions = 2^{2^n}
Total self dual functions = $2^{2^{n-1}}$

- ① K Map:- It is a graphical representation for simplifying the boolean algebra.
- ② n variable boolean expression contains 2^n no. of cells in K Maps.
- ③ K MAP is based on Gray code (also known as unit distance code)

④ K MAP is based on 3 type values, i.e. 0, 1 and ϕ

002
012
112
102
change



$f(A, B, C)$

AB	00	01	11	10
00	0	0	1	1
01	0	1	1	1
11	0	1	1	1
10	0	1	1	1

$\Sigma (0, 2, 5, 7, 9, 11, 13, 15)$

$$= \bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB$$

$$= \bar{A}(\bar{B} + B) + A(\bar{B} + B)$$

$$= \bar{A}(1) + A(1)$$

$$= \bar{A} + A$$

$$= 1$$

f	f
0	1
1	0
0	0
0	0

$\bar{A}\bar{B}\bar{C}, \bar{A}B, A\bar{B}, A+B, \bar{A}+\bar{B}, A+B, \bar{A}+\bar{B}$

$= 2^n$
in
 $n = 2^2$
 $(n-1)$

al representation
clean algebra.

in k maps

code (
 code)

④ K MAP is based on 3 type of input values, i.e 0, 1 and don't care.

0 0 2 bit and 4 bit
0 1 2
1 1 2
1 0 2
change

A \ BC	00	01	11	10
0	0	1	1	2
1	4	5	7	6

f(A,B,C)

A \ BC	00	01	11	10
0	0	1	1	2
1	4	5	7	6

$\Sigma (0, 2, 5, 7, 2, 9, 11)$
 $\Sigma (3, 8, 10, 2, 12, 14)$

$$= \bar{A}\bar{B} + \bar{D}\bar{A}B + \bar{A}\bar{B}\bar{D}$$

$$= \bar{B} (A + \bar{A}\bar{D}) + D\bar{A}B$$

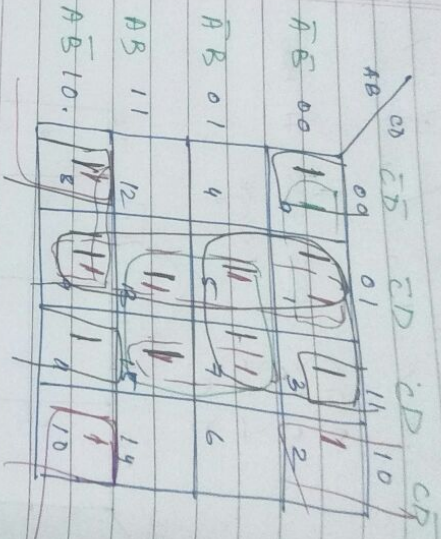
$$= \bar{B} (A + \bar{A}) (A + \bar{D}) + D\bar{A}B$$

$$= \bar{B} A + \bar{B} \bar{A} + \bar{D} A + D\bar{A}B$$

24/10/28

Classwork

Q. 5:



$$\bar{C}\bar{D} + A\bar{B}\bar{D} + BCD + C\bar{D}\bar{B} +$$

$$A\bar{B}\bar{C}$$

$$A\bar{B}\bar{C} + \bar{C}\bar{D} + B\bar{D} +$$

$$\sum m = 0, 1, 3, 5, 7, 8, 9, 11, 13, 15$$

$$\bar{A}\bar{D} + \bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D} + C\bar{D}\bar{A}$$

$$= \bar{D} + \bar{C}\bar{D}\bar{B} + (\bar{D} + \bar{D})(\bar{D} + \bar{B}\bar{C})$$

$$= \bar{D} + \bar{B}\bar{C}$$

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